Composition is a nerve-level-two datum

Grading compositional invariants of directed containers by nerve level, not cohomological degree — a program, with proved instances

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8 October 2026; revised 10 October 2026; corrected 11 October 2026 (Rick round-2: A1, A2, A3); revised 13 October 2026 (Rick 2026-10-13: Sec 2 and Rem 13 reconciled with Rem 17 — it is E_ω, not $\mathcal{S}$, that splits)

supersedes content commit `6089268`; content commit `ccbf3b6`

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Abstract

A directed container is a small category, and a question about such a category can be a question about its underlying shape, about a resource transported across it, or about how its morphisms compose. We ask whether a numerical grade on an invariant records how much of composition the invariant can see, and we argue that the right grade is the *nerve level* the invariant reads, not — as an earlier note of ours claimed — its cohomological degree. The two indices differ by one: a datum on composable pairs lives at nerve level two, but its cohomology is first read by a degree-*one* cocycle condition. Out-degree homogeneity reads only the reflexive graph (nerve level one) and is provably blind to composition. The comultiplication δ is a nerve-level-two datum, and the invariants built from it — the left-translation trivialization of the tangent bundle, detecting groupoids and, one notch finer, thin groupoids — separate the group $\mathbb{Z}/2$ from the idempotent monoid $\{1, e\}$ at cohomological degree *one*, exactly the separation the floor cannot make. The Zappa–Szép weld $[\omega] \in H^2$ is a nerve-level-*three* datum whose defining cocycle *condition* is associativity, a predicate on composable triples; the cohomology *class* itself measures not associativity but whether an already-associative weld *splits* into its claimed factorization. We prove that the truncation tower collapses at level two (a category is its 2-truncated nerve), so above the floor there is no third *detection* rung; the genuine level-two/level-three distinction is the condition-level fact that building a category from a composition is an associativity problem. Finally we show the degree-one inventory class $[\theta_R]$ and the degree-two weld class $[\omega]$ are two independent anchors in the composition-sensitive band, differing in coefficients and condition-level, not in whether they see composition. We assemble these as a program — nerve level grades compositional data — and flag, as an open question with a degree-one counterexample, whether cohomological *degree* stratifies *detection* above the floor.

*Kodamai. This is a positioning note for the Kodamai AI-Mathematician programme; the proofs it organizes are established in the cited companion notes and proof files, which live together with this note and the verification scripts at `github.com/scotmacbeth/work-in-progress` (content commit recorded on this page). It *supersedes and corrects* the earlier note *"Composition is a degree-two datum"* (content commit `ae9d67a`), whose headline grading was off by one; see the Introduction. Prepared as a framing appendix for the Strathclyde mini-course *Tangent Categories and Containers* (13–16 October 2026).

1 Introduction

A correction, stated first. An earlier note of ours carried the headline *"Composition is a degree-two datum,"* and with it the claim that *no invariant below cohomological degree two can detect composition.* The claim is false, and this note replaces it. The note's own degree-*one* invariant already separates the group $\mathbb{Z}/2$ from the idempotent monoid $\{1, e\}$ — a pair that differs only in how its one nontrivial arrow composes with itself. The error was a conflation of two distinct simplicial indices attached to a cohomology class, and the fix is to grade by the right one. The underlying theorems are unaffected; only the grading, and the slogan that named it, were wrong. We state the corrected grading, prove the regrade that forces it, and rebuild the ladder honestly.

One functor, three questions. A directed container is, by the theorem of Ahman and Uustalu [2], exactly a small category: the comonoids for the composition product $\lhd$ on polynomial functors are small categories, with objects the shapes and, for each object a, the direction set E_a as the set of morphisms out of a. A single such category carries structure at several levels at once. There is its *underlying functor* $p = \sum_a y^{E_a}$, which remembers only the object-indexed family of direction sets. There is *data transported across it* — a presheaf of resources, say stock levels at the nodes of a supply chain. And there is its *composition* — the comultiplication $\delta\colon p \to p \lhd p$, which records how two morphisms compose, and, for a composite built as a Zappa–Szép product, the weld that glues two factors together. Questions can interrogate any of these levels, and a recurring experience of this programme is that an invariant designed for one is *structurally unable* to answer another. The derivative is blind to composition; the consistency of a transported resource is independent of whether the composite factors as claimed. These are facts about where the relevant datum lives.

The two indices, and the regrade. Make that experience precise by grading an invariant by the *simplicial level of the nerve* the datum it reads occupies. The nerve $\mathrm{N}(\mathcal{C})$ of a category has N_0 the objects, N_1 the arrows, and N_k the composable k-tuples; composition is the inner face $d_1\colon \mathrm{N}_2 \to \mathrm{N}_1$, so the comultiplication δ is a datum on N_2, the composable pairs. This is the "two" that the old slogan heard. But the *cohomological degree* of a class is a different index, and it is lower by one:

> *An n-cochain is a function on N_n, but the cocycle condition $d^n c = 0$ that defines its class is a system of equations indexed by N_{n+1}. The binding simplicial level of an H^n-class is thus N_{n+1}: cohomological degree = (nerve level of the cocycle condition) -1.*

Composable pairs (N_2) are therefore read already by a degree-*one* cocycle condition. The old "degree two" took the nerve level of δ and mislabelled it as a cohomological degree. We correct this throughout (§2, Lemma 1); the retired slogan becomes:

> *Composition — the datum of composable pairs — is a nerve-level-two datum, first detectable at cohomological degree one. Associativity — a* condition *on composable triples — is the nerve-level-three cocycle condition, cohomological degree two; the Zappa–Szép weld class $[\omega] \in H^2$ lives there, but as the obstruction to* splitting *an already-associative weld, not as a test of associativity itself.*

The honest ladder. The detection ladder has, above the floor, *two* rungs for the price the old note charged for three, and a genuine third rung that is not a detection rung at all.

- **Nerve level 1 (the blind floor).** Invariants that factor through the reflexive graph $\mathrm{tr}_1\mathrm{N}(\mathcal{C})$ read nothing about composition — and those factoring through the strictly coarser forgetful $U\colon \mathsf{Cat} \to \mathsf{Poly}$, such as tangent-regularity, read only the out-degree family. Tangent-regularity is provably blind: $\mathbb{Z}/2$ and $\{1, e\}$ have the same underlying functor y^2 and are indistinguishable to it.

- **Nerve level 2 (composition).** Invariants that read the composition map $d_1\colon \mathrm{N}_2 \to \mathrm{N}_1$ can see composition. Two live here. The δ-translation trivialization detects groupoids, and one notch finer thin groupoids, and it *separates* $\mathbb{Z}/2$ *from* $\{1, e\}$ — *at cohomological degree one.* The inventory class $[\theta_R] \in H^1(\mathcal{S}; A)$ of a transported resource is the other.

- **Nerve level 3 (associativity).** A *condition* on composable triples — associativity — lives one level above composition; its honest witness is a non-associative unital magma (Theorem 12), a pointwise N_3 failure. The Zappa–Szép weld $[\omega] \in H^2$ is the degree-two cohomology class at this level: its defining *cocycle condition* is associativity, but the class itself measures whether an already-associative weld *splits* into its claimed factorization. For a *fixed* category this rung is empty of new detection power — a category is reconstructible from its 2-truncated nerve (Theorem 11), so the truncation tower collapses at level two. The genuine level-two/level-three distinction is in the *assembly* problem, where the object varies: composition exists (N_2) versus composition is associative (N_3).

This is a program, not a theorem. What is proved is a regrade lemma, a strict floor separation, a collapse theorem, an assembly separation, three invariant instances, and one independence result — each in a companion note or proof file cited below and not re-derived here. What is *not* proved, and what we are careful to state as an open question (§7), is that cohomological *degree* stratifies *detection* power above the floor. As stated naively ("higher degree detects strictly more") this is **false**: the degree-one δ-invariant already detects composition, and the degree-one and degree-two cohomological anchors are *independent*, not ordered. The grade that does the stratifying is *nerve level*; cohomological degree is a coefficient-dependent shadow of it (Remark 8). This is a positioning note: its job is to make the pattern visible and correctly labelled, not to establish new mathematics.

Outline. Section 2 fixes containers as categories, the nerve and its two truncations, the regrade lemma, the two structure maps ∂ and δ, the resource torsor and its H^1, the weld and its H^2, and the nerve-level grading. Section 3 is the blind floor; Section 4 is composition, where the δ-translations separate what the floor cannot and the monodromy tower stratifies them; Section 5 is associativity, the weld, and the collapse/assembly facts; Section 6 shows the degree-one inventory class and the degree-two weld class are independent. Section 7 states the program and the open question.

Note on this revision (11 October 2026). Following Rick's round-2 report we sharpen the splitting derivation of §5 without changing any theorem. **A1:** the cocycle condition $d\omega = 0$ is associativity of the *twisted composition over the already-associative base* $\mathcal{S}$, *not* the base-associativity equation of Theorem 12 (a magma, over which $d \circ d \neq 0$ and no class exists); and, by [12], $[\omega] = 0 \Leftrightarrow (G)$ while *splitting* is $(L) \wedge (G)$, so splitting is decided by $[\omega]$ only under the standing local condition (L) (Remark 13). **A2:** $[\omega] \in H^2(\mathcal{S}; D)$ is a class of the *base* $\mathcal{S}$ classifying an extension $D \rightarrowtail E_\omega \twoheadrightarrow \mathcal{S}$; it is E_ω that splits, which is why a base presented as $\mathcal{C} \bowtie \mathcal{D}$ has $[\omega] = 0$ while the rigid-twist witness ($\mathcal{S} = B(\mathbb{Z}/2)$, $D = \mathbb{Z}/2$, $E_\omega = B(\mathbb{Z}/4)$) has $[\omega] \neq 0$ (Remark 17). **A3:** under the regrade convention (nerve level = degree $+1$), only the binding *datum* $a \cdot a = 1$ is coefficient-free at N_2; the detecting class sits at N_2 over $\mathbb{F}_2$ and N_3 over $\mathbb{Z}$ (Remark 8).

2 Preliminaries

We work with polynomial functors in one variable and finite support; everything is first order.

Containers as categories. A *container* in one variable is a polynomial functor $p = \sum_{a \in \mathrm{Ob}} y^{E_a}$ with a set Ob of *shapes* and, for each shape a, a set E_a of *directions* [1, 3]. The composition product $\lhd$ has unit y and realizes substitution. By Ahman and Uustalu [2] the comonoids in $(\mathsf{Poly}, \lhd, y)$ are exactly small categories: a comonoid structure $\delta\colon p \to p \lhd p$ chooses identities (the counit) and a composition (the comultiplication), Ob is the object set, and $E_a = \coprod_c \mathrm{Hom}(a, c)$ is the set of morphisms out of a, so $|E_a|$ is the *out-degree* of a. Write $\mathsf{Comon}(\mathsf{Poly}, \lhd) \simeq \mathsf{Cat}$ for this equivalence and $U\colon \mathsf{Cat} \to \mathsf{Poly}$ for the forgetful functor that discards δ and keeps only $\sum_a y^{E_a}$. We compose *diagrammatically*: for $g\colon a \to b$ and $h\colon b \to c$, the composite $g \cdot h\colon a \to c$ means "g then h".

The nerve and its two truncations. The *nerve* $\mathrm{N}(\mathcal{C})$ is the simplicial set with $\mathrm{N}_0 = \mathrm{Ob}$, N_1 the arrows, and N_k the composable k-tuples $a_0 \xrightarrow{f_1} \cdots \xrightarrow{f_k} a_k$. The inner face $d_1\colon \mathrm{N}_2 \to \mathrm{N}_1$ composes the pair, $(f, g) \mapsto f \cdot g$; the degeneracies insert identities. The *k-truncation* $\mathrm{tr}_k\mathrm{N}(\mathcal{C})$ keeps $\mathrm{N}_0, \ldots, \mathrm{N}_k$ with all faces and degeneracies among them. Two truncations matter here:

$$\mathrm{tr}_1\mathrm{N}(\mathcal{C}) = \text{the reflexive graph of } \mathcal{C} \text{ (objects, arrows, source, target, identities);}$$

$\mathrm{tr}_2\mathrm{N}(\mathcal{C})$ adds the composition map d_1. The underlying functor $U(p) = \sum_a y^{E_a}$ is a *strictly coarser* shadow of $\mathrm{tr}_1\mathrm{N}(\mathcal{C})$, not an equal: it records only the out-degree family $a \mapsto |E_a|$ and forgets which arrow runs where. For instance the interval category $[1] = (0 \to 1)$ and the disjoint union $B(\mathbb{Z}/2) \sqcup \mathrm{pt}$ both have $U = y^2 + y$ yet have non-isomorphic reflexive graphs (an arrow between two objects versus a loop at one). Hence an invariant factoring through U is blinder even than one reading all of $\mathrm{tr}_1\mathrm{N}(\mathcal{C})$; the floor of our ladder (§3) sits at this coarsest reading. For an abelian coefficient system A (a functor $\mathcal{C} \to \mathsf{Ab}$; constant unless noted), $H^n(\mathcal{C}; A)$ is the cohomology of the classifying space $B\mathcal{C}$, computed by the normalized cochain complex with $C^n = \{\text{functions on nondegenerate } n\text{-simplices}\}$ and the standard bar differential; for a one-object category it is monoid/group cohomology.

Two simplicial indices, and the regrade lemma. Two different simplicial levels attach to a cohomological datum, and conflating them is the kind of error this note exists to correct; we name them once and keep them apart. The *cochain level* of a datum is the k for which it is a function on the k-simplices N_k: the comultiplication δ and any bar n-cochain have cochain level equal to the number of arrows they take as input. The *nerve level* of an invariant — the grade this note uses throughout — is instead the highest N_k entering the *definition* of the invariant's class: for a class in H^n this is the level N_{n+1} that indexes its cocycle *condition*, and for a raw datum presented on N_k it is k. Cohomological degree is a third, strictly lower index. For δ the two simplicial indices coincide (both are 2); for a cohomology class they diverge, and the divergence is exactly one — as the following lemma records.

Lemma 1 (regrade)**.** *For every $n \geq 0$, the cocycle condition $d^n c = 0$ defining an H^n-class is a system of equations indexed by N_{n+1}. Consequently*

$$\textit{cohomological degree } n \;=\; (\textit{nerve level of the cocycle condition}) \;-\; 1.$$

*In particular a datum on composable pairs (N_2) is read by a degree-*one *cocycle condition, and one on composable triples (N_3) by a degree-two cocycle condition.*

Reason. $d^n\colon C^n \to C^{n+1}$ sends a function on N_n to a function on N_{n+1}: by the bar formula $(d^n c)(f_1,\dots,f_{n+1})$ is an alternating sum of c on the n-faces of the $(n+1)$-simplex $(f_1,\dots,f_{n+1})$, together with the coefficient action along f_1. Hence $d^n c = 0$ is the family $\{(d^n c)(\sigma) = 0 : \sigma \in \mathrm{N}_{n+1}\}$, indexed by N_{n+1}; the coboundary relation reads only N_n, so the highest level entering the definition of the class is N_{n+1}. Proved in full in [7]. □

Remark 2 (the "two" of $p{\lhd}p$, and a warning about the weld)**.** The comultiplication $\delta\colon p \to p{\lhd}p$ lands in the two-fold composite $p{\lhd}p$: it is data on composable *pairs*, i.e. it has *cochain level* two, and in that sense the old note's "two" was heard correctly. Its error was to call that two a *cohomological degree*: by Lemma 1 the H^1 cocycle condition of a datum on 2-simplices already reads them, so composition is nerve level two but cohomological degree *one*. The same lemma issues a warning that matters below: cochain level and nerve level need not agree. The weld $[\omega] \in H^2$ is a 2-cochain — cochain level two, exactly like δ — yet its cocycle condition is indexed by N_3, so its nerve level is three. Cochain level thus fails to separate the weld from composition; nerve level separates them. Every "degree two" of the earlier note that referred to composition itself should read "nerve level two, cohomological degree one".

Two structure maps: ∂ reads the graph, δ reads composition. The one-hole derivative [1] is

$$\partial p \;=\; \sum_a |E_a| \cdot y^{|E_a|-1},$$

depending only on the direction sets, i.e. only on $U(p)$: a construction on the out-degree family — the coarsest floor reading, coarser even than the reflexive graph $\mathrm{tr}_1\mathrm{N}(\mathcal{C})$ — so a nerve-level-one construction. With $W = \mathbb{Z}[\varepsilon]/(\varepsilon^2)$ the dual-number Weil algebra, the attached tangent functor is the Weil base change $T = (-)\otimes W$; the one-hole derivative appears as the ε-coefficient of the substitution $p(y+\varepsilon v)$, and all we use downstream is that the vertical fibre over a shape a is free of rank $|E_a|$ [14, 8]. By contrast $\delta\colon p \to p{\lhd}p$ is *not* a construction on $U(p)$: it is the composition map d_1, a datum on N_2. This is the precise sense in which composition is a nerve-level-two datum, and the datum ∂ cannot see.

The resource torsor and its degree-one class. Model a supply chain as a small category $\mathcal{S}$: objects are nodes, morphisms out of a are admissible operations, composition is sequencing. *Inventory* is data transported across $\mathcal{S}$: a local system in which each operation f acts on stock by a translation $R(f)(x) = x + \tau_f$ of an abelian stock-group A. (We write τ_f for the translation 1-cochain to avoid any clash with the comultiplication δ.) A *global section* is a node-potential φ with

$$\tau_f \;=\; \varphi(\mathrm{dom}\, f) - \varphi(\mathrm{cod}\, f) \qquad \text{for every } f,$$

i.e. the 1-cochain τ is a coboundary. Because the translations are not group homomorphisms, the fibrewise data form a *torsor* under the linear local system A, not a presheaf of groups; the obstruction is accordingly the class

$$[\theta_R] \;:=\; [\tau] \in H^1(\mathcal{S}; A).$$

This is a degree-one class, coefficient system the stock-group A, measuring route/reconvergence discrepancy over the nerve of $\mathcal{S}$ — the classical mechanism of Kirchhoff's voltage law (§6).

The weld and its degree-two class. A composite chain is *assembled* from procurement $\mathcal{C}$ and logistics $\mathcal{D}$; the target structure is a *Zappa–Szép (matched-pair) product* $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$, in which $\mathcal{C}, \mathcal{D}$ are wide subcategories meeting in the discrete category on the shared objects and every morphism factors uniquely as $f = c \cdot d$ [2, 12]. Whether an *associative* candidate weld (one satisfying the matched-pair/cocycle condition of §5) *splits* is governed by the *weld obstruction*

$$[\omega] \in H^2(\mathcal{S}; D),$$

a Baues–Wirsching 2-cocycle on the orbit category with coefficients the vertex-group natural system D (the one determined by the candidate normal factor, not the wide subcategory $\mathcal{D}$ itself). Precisely, $[\omega]$ classifies an extension $D \rightarrowtail E_\omega \twoheadrightarrow \mathcal{S}$ of the base $\mathcal{S}$: the chain satisfies the global-closure condition (G) exactly when $[\omega] = 0$, and then — given also the local condition (L) — it is the extension E_ω (equivalently, the candidate weld) that *splits*, realising the matched-pair factorization $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ [12]. It is E_ω, not $\mathcal{S}$ itself, that splits: a base *presented* as $\mathcal{C} \bowtie \mathcal{D}$ has $[\omega] = 0$ by construction (Remark 17). Two simplicial facts about this class must be kept apart (§5). Its defining cocycle *condition* is the associativity equation of the assembled composition, a system on composable triples N_3 — so by Lemma 1 the class is nerve level three, cohomological degree two. The *class* itself, however, presupposes that condition: a cochain that violates it is not a weld at all, and there is then no class to form. What $[\omega]$ measures is accordingly not whether the weld is associative but whether an *associative* weld *splits* — the Eilenberg–Mac Lane classification of the extension. It is the home of the blockchain re-entrancy obstruction $[\omega] = \varepsilon \in H^2 \cong \mathbb{Z}/2$, which is therefore a non-split but associative weld, not a non-associative one [13].

The nerve-level grading. We rank a compositional invariant by the nerve level of the binding datum it reads: level 1 for the out-degree family $U(p)$ (the coarsest floor reading, below even the reflexive graph $\mathrm{tr}_1\mathrm{N}$); level 2 for the composition map d_1 — read by the comultiplication δ and by any $H^1(\mathcal{S}; A)$ of a transported local system; level 3 for associativity of the assembled composition, a condition on composable triples whose honest witness is a non-associative magma and at which the Zappa–Szép weld $[\omega] \in H^2$ sits as the splitting obstruction of an associative weld. The grade is a grade of *data*; the different cohomological degrees carry different coefficient systems, and as Remark 18 records, the same cyclic group can surface at different degrees under a change of coefficients, so one must not read the grading as a single filtration.

3 Nerve level one: the invariant blind to composition

The floor reads only the reflexive graph and is for that reason unable to see composition at all. The result is proved in [8]; we state it and exhibit the discriminating examples.

The tangent bundle $Tp = p \otimes W$ has, over a shape a, a fibre free of rank $|E_a|$. Call p *tangent-regular* if these fibres are all isomorphic — the honest residue, in the representable/monoidal setting, of parallelizability in the sense of Cruttwell, Ikonicoff, Lemay and Van Der Linden [4, Def. 4.18, attributed there].

Theorem 3 ([8])**.** *A container $p = \sum_a y^{E_a}$ is tangent-regular iff the out-degree $a \mapsto |E_a|$ is constant on* Ob*. This condition depends only on $U(p)$ — it factors through the forgetful $U\colon \mathsf{Cat} \to \mathsf{Poly}$, hence through a reading even coarser than the reflexive graph $\mathrm{tr}_1\mathrm{N}(\mathcal{C})$ — and is therefore blind to composition: it is invariant under changing δ on a fixed underlying functor p, and so cannot be equivalent to any property that distinguishes two comonoid structures on the same p, such as being a groupoid (Example 4).*

Example 4 (the blindness, made concrete)**.** The idempotent monoid $\{1, e\}$ with $e^2 = e$ and the group $\mathbb{Z}/2$ both have underlying functor y^2. They have the *same* tangent bundle and are both tangent-regular (one object, so out-degree is trivially constant), yet $\mathbb{Z}/2$ is a groupoid and $\{1, e\}$ is not. The floor cannot tell them apart. Conversely the groupoid $\mathbb{Z}/2 \sqcup \mathbb{Z}/3$ has out-degrees 2 and 3, so $U = y^2 + y^3$ is *not* tangent-regular though it is a groupoid: tangent-regularity neither implies nor is implied by being a groupoid.

The cautionary form of this rung is easy to fall for. A genuine theorem of Lanfranchi [4, Thm. 4.20] — *for a group object G in a Cartesian tangent category, G is a Lie group object iff G is parallelizable* — invites the port "a container is parallelizable iff it is a groupoid." The hypothesis "given a group object" is essential: without it the statement is false (S^7 is parallelizable with no group structure), and the port drops it. The port is in any case ill-posed, because the container derivative's tangent structure is the monoidal base change $(-)\otimes W$ and not a Cartesian one [8]; what survives is tangent-regularity, which lives at nerve level one and cannot see the groupoid property. This is no defect in [4] — that theorem is about group objects in Cartesian tangent categories — but the symptom of asking a graph-level invariant a composition-level question. Connected groupoids *are* tangent-regular; but strictly more categories are, and the gap is the composition content that only nerve level two recovers (§4).

4 Nerve level two: composition, and the invariant that sees it — at degree one

The second rung reads the composition map d_1. Here composition becomes visible, and — this is the crux of the correction — it becomes visible already at cohomological degree *one*. We first exhibit the δ-translation invariant that separates the pair the floor could not, then verify by direct computation that the separation is a degree-one cohomological fact, and finally refine the invariant into a monodromy tower.

The δ-translation invariant. Fix a morphism $g\colon a \to b$ and pin the first slot of δ to it: the *δ-translation*

$$L_g\colon E_b \to E_a, \qquad L_g(h) = g \cdot h,$$

precomposition by g. It is contravariantly functorial, assembling δ into the *arrows-out presheaf* $E_{(-)}\colon \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$, $a \mapsto E_a$, $g \mapsto L_g$ — a property-free restatement of the composition map d_1. (The statement below concerns this vertical/linear layer of $Tp = p \otimes W$, not the full bundle [10].) Say Tp is *δ-trivializable* over a connected component $[a]$ if $E_{(-)}$ restricted there is a *local system* — every L_g a bijection — the honest discrete analogue of trivializing a tangent bundle by translations, $TG \cong G \times \mathfrak{g}$.

Theorem 5 ([10])**.** *Tp is δ-trivializable over $[a]$ iff the full subcategory on $[a]$ is a* groupoid*; globally, iff $\mathcal{C}$ is a groupoid. Surjectivity of the translations already forces full invertibility: a surjective L_g hits $\mathrm{id}_a \in E_a$, so some h has $g \cdot h = \mathrm{id}_a$; surjectivity of L_h then gives k with $h \cdot k = \mathrm{id}_b$, whence $g = g \cdot (h \cdot k) = (g \cdot h) \cdot k = k$ and so $h \cdot g = \mathrm{id}_b$ — g is invertible. The dividing line is exactly "groupoid", with no weaker cancellation condition between.*

Example 6 (seeing what the floor could not)**.** Return to the pair of Example 4. On $\mathbb{Z}/2$ every L_g is a bijection, so Tp is δ-trivializable. On $\{1, e\}$ the translation L_e sends $1 \mapsto e$ and $e \mapsto e$, missing $\mathrm{id} = 1$; it is not even injective, and $\{1, e\}$ is not δ-trivializable. The invariant *separates $\mathbb{Z}/2$ from $\{1, e\}$* — the exact pair the floor, reading only the common underlying functor y^2, was structurally

unable to tell apart (Theorem 3). This is the content of the slogan we can prove: there is a genuine invariant, built from δ (the composition map d_1), that detects a composition property no invariant factoring through U can.

The separation is a degree-one fact. It remains to pin the cohomological degree at which this separation occurs, and to show it is *one*, not two. This is Theorem 2 of [7]; we record the computation because it is where the old slogan broke.

Theorem 7 ([7])**.** *The categories* $\mathbb{Z}/2$ *and* $\{1, e\}$ *have isomorphic* 1*-truncated nerves (hence agree on every invariant factoring through* U*, in particular on out-degree), but* $\dim_{\mathbb{F}_2} H^1(-;\mathbb{F}_2)$ *distinguishes them:* $H^1(\mathbb{Z}/2;\mathbb{F}_2) = \mathbb{F}_2$ *while* $H^1(\{1,e\};\mathbb{F}_2) = 0$*. The distinction is a function of* $\mathrm{tr}_2\mathrm{N}$ *and not of* $\mathrm{tr}_1\mathrm{N}$*. Hence detection of composition begins at nerve level two, which by Lemma 1 is cohomological degree* one*.*

Computation. The unique nondegenerate composable pair is (a, a) with $d_1(a, a) = a \cdot a$, which is 1 in $\mathbb{Z}/2$ and a in $\{1, e\}$; so no isomorphism of tr_2 nerves exists, while the relabelling of $\{1, a\}$ is an isomorphism of tr_1 nerves. Over $\mathbb{F}_2$ a normalized 1-cochain is a value $c(a)$. For $\{1, e\}$: $(d^1c)(e, e) = c(e) - c(e \cdot e) + c(e) = c(e)$, forcing $c(e) = 0$, so $H^1 = 0$. For $\mathbb{Z}/2$: $(d^1c)(a, a) = 2c(a) - c(1) = 0$ over $\mathbb{F}_2$, so every value is a cocycle and $H^1 = \mathbb{F}_2$. Full details and a verifying script in [7]. □

Remark 8 (why *nerve level*, not degree, is the grade — the coefficient test)**.** The degree at which this single N_2 distinction surfaces *depends on the coefficients.* The feature $a \cdot a = 1$ creates a 1-cycle, $H_1(\mathbb{Z}/2) = \mathbb{Z}/2$, and is a raw *datum* on nerve level N_2. Over $\mathbb{F}_2$ it is detected by H^1 (degree one); over $\mathbb{Z}$ the same torsion is detected by H^2 (degree two, $H^2(\mathbb{Z}/2;\mathbb{Z}) = \mathbb{Z}/2$), the Bockstein having moved it up a degree. So "which degree detects composition here" has no coefficient-free answer. What *is* coefficient-free is the nerve level of the *binding datum*: $a \cdot a = 1$ sits on N_2 for *every* coefficient ring. The *detecting class*, by contrast, obeys the regrade convention (nerve level = degree +1), so it sits at N_2 over $\mathbb{F}_2$ (where it is H^1) but at N_3 over $\mathbb{Z}$ (where it is H^2). The thesis is therefore a statement about the datum, not the detecting class — and it is also why the old "degree two" was not even stably wrong: over $\mathbb{F}_2$ the detection is degree one.

A monodromy tower on the δ-translations. The δ-translations can be asked to do more than be invertible: to trivialize to a *constant* identification, with trivial monodromy. Call $E_{(-)}$ *constant-trivializable* over $[a]$ if it is naturally isomorphic to a constant presheaf Δ_V: bijections $\varphi_b\colon E_b \to V$ with $\varphi_x \circ L_g = \varphi_y$ for *every* $g\colon x \to y$, loops included. Read against a loop this asks the translation cocycle $\{L_g\}$ to be a coboundary.

Theorem 9 ([11])**.** $E_{(-)}$ *is constant-trivializable over* $[a]$ *iff* $[a]$ *is a* thin *(indiscrete) groupoid — trivial vertex groups, i.e. an equivalence-relation component; globally, iff* $\mathcal{C}$ *is a thin groupoid. Constant-trivializability implies* δ*-trivializability, so the groupoid hypothesis of Theorem 5 comes for free; applying naturality to a loop* $g \in \mathrm{Aut}(a)$ *forces* $L_g = \mathrm{id}$*, and since the regular action fixes the basepoint* $\mathrm{id}_a \in E_a$*, forces* $g = \mathrm{id}_a$ *— the vertex group vanishes.*

Corollary 10 (the monodromy tower)**.** *Over a connected component, three successively finer conditions on the* δ*-translation presheaf* $E_{(-)}$ *stratify the composition datum:*

Rung	*Condition on* $E_{(-)}\big\vert_{[a]}$	*Equivalent to*	*Status*
1	*generic presheaf (each* L_g *a function)*	*nothing*	*trivial*
2	*local system (every* L_g *a bijection)*	$[a]$ *a groupoid*	*[10]*
3	$\cong$ *constant presheaf*	$[a]$ *a thin groupoid*	*[11]*

Rungs 2 *and* 3 *are distinct, separated by a nontrivial vertex group:* $\mathbb{Z}/2$ *is a groupoid but not constant-trivializable (the regular action* $\mathbb{Z}/2 \curvearrowright \mathbb{Z}/2$ *is free, nontrivial loop monodromy), while the indiscrete two-object category* I_2 *is both. The obstruction to upgrading rung* 2 *to rung* 3 *is the* class of the regular representation *in the non-abelian* $H^1([a]; \Sigma_{E_a})$*; this class is* $\mathrm{Hom}(\mathrm{Aut}(a), \Sigma_{E_a})/\mathit{conj}$ *evaluated at the regular representation, which is faithful, so it vanishes iff* $\mathrm{Aut}(a) = 1$*. Thus* $\mathrm{Aut}(a)$ *is the* holonomy group *of the tower, and it is a nontrivial* vertex group *— not one-object-ness — that marks the step.*

So the composition datum carries an internal grading of its own: generic $\subsetneq$ groupoid $\subsetneq$ thin groupoid, all three living at nerve level two as refinements of the δ-translation presheaf. Rung 3 has no classical counterpart among positive-dimensional Lie groups — every Lie group already admits a global flat left-invariant framing with trivial holonomy — because in the discrete setting rung 3 forces the group acting on the fibre to be trivial. We do not overclaim: rungs 2 and 3 are composition-sensitive invariants of δ, and the literal cohomological anchor of rung 3 is the non-abelian degree-one class above, not a new H^2 class. The literal abelian anchors of the ladder remain $[\theta_R] \in H^1$ and $[\omega] \in H^2$; the tower refines the *datum* δ at nerve level two.

5 Nerve level three: associativity, and the weld as splitting obstruction

The predicates one can impose on a composition climb one more level, from "a composition exists" on N_2 to "it is associative" on N_3. By Lemma 1 the degree-two cocycle *condition* of the weld $[\omega] \in H^2$ is a system of equations on composable triples N_3, and that system is exactly associativity — though, as we will see, the *class* $[\omega]$ measures splitting rather than associativity itself. We make the level-two/level-three distinction precise, and show why it is *not* a third detection rung for a fixed category.

Theorem 11 (2-coskeletal collapse, [7])**.** *A small category is determined up to isomorphism by its* 2*-truncated nerve. Consequently, for isomorphism-invariants of a fixed category, every invariant factors through* $\mathrm{tr}_2\mathrm{N}$*: there is no invariant that needs* N_3*, and the truncation tower stops at level two.*

Reason. From $\mathrm{tr}_2\mathrm{N}(\mathcal{C})$ one recovers objects, arrows, source, target, identities, and composition $d_1 \colon \mathrm{N}_2 \to \mathrm{N}_1$; these are all the data of a category, and an isomorphism of 2-truncated nerves is exactly an isomorphism of categories. Higher N_k are forced by the Segal condition. See [7]. □

So the third rung is not a detection separation of *fixed* categories. It is real, but it lives in the *assembly* problem, where the object varies: one is handed a composition on composable pairs and asks whether it assembles into a category.

Theorem 12 (associativity is an N_3-predicate, [7])**.** *There is a unital magma whose full composition on composable pairs is defined and unital, yet which is not associative — e.g. on* $\{1, x, y\}$ *with unit* 1 *and* $x \cdot x = y$*,* $x \cdot y = x$*,* $y \cdot x = y$*,* $y \cdot y = y$*: the triple* (x, x, x) *gives* $(x \cdot x) \cdot x = y$ *but* $x \cdot (x \cdot x) = x$*. Every pair-level (*N_2*) datum is present and unital; the obstruction lives purely at* N_3*. The separation of "composition exists" (*N_2*) from "composition is associative" (*N_3*) is therefore strict, and its witness is a pointwise* N_3 *condition failure — not a cohomology class.*

Remark 13 (the weld is a splitting obstruction, not an associativity test)**.** It is tempting to read the Zappa–Szép weld class $[\omega] \in H^2$ as the cohomological form of this non-associativity. That reading

is wrong, and the regrade makes the error visible. The cocycle *condition* $d\omega = 0$ *is* associativity of the *twisted composition over the already-associative base* $\mathcal{S}$ — *not* the associativity equation of Theorem 12. The two are distinct N_3-predicates on *different objects*: Theorem 12 is a failure of associativity of a *base* composition (a magma), over which there is no cochain complex at all ($d \circ d \neq 0$, so there is no H^2 for a class to live in), whereas $d\omega = 0$ is a condition on a cochain over a base that is *already* a category. A cochain that fails $d\omega = 0$ is not a weld, and there is then no class. A class $[\omega]$ therefore presupposes associativity; what it classifies, by Eilenberg–Mac Lane and in the general matched-pair form of [12], is whether the associative weld satisfies the global closure condition (G), with $[\omega] = 0 \Leftrightarrow (G)$; given also the local condition (L), (G) is exactly the splitting of the extension E_ω — equivalently, the candidate weld then realises the matched-pair factorization $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$. It is E_ω, not $\mathcal{S}$ itself, that splits (Remark 17); so under the standing (L) the class decides splitting [12]. (Eilenberg–Mac Lane alone covers the *semidirect* case $\mathcal{D} \rtimes \mathcal{C}$; the general matched-pair statement, where neither factor need be normal, is the theorem of [12], classified cohomologically after Baues–Wirsching.) The blockchain re-entrancy obstruction $[\omega] = \varepsilon \neq 0 \in H^2 \cong \mathbb{Z}/2$ [13] is accordingly a *non-split but associative* weld, not a non-associative one: the cocycle condition is met, so associativity holds; what fails is the decomposition. Both phenomena are nerve-level-three — the non-associativity witness a pointwise condition on N_3, the weld a class whose cocycle condition lives on N_3 — but they measure different things, and only the first is a failure of associativity.

This is the honest content of "nerve level two $\subsetneq$ nerve level three": not that a finer invariant of a fixed category needs triples — Theorem 11 forbids that — but that the predicates one can impose on a composition climb a genuine level, from "exists" on N_2 to "associative" on N_3. The weld $[\omega] \in H^2$ earns its place at level three through its cocycle condition, which lives on N_3 (Remark 13); it is the programme's level-three *cohomological* anchor, measuring splitting, while associativity itself is the level-three *condition*, witnessed pointwise by the magma.

6 The two cohomological anchors are independent

The ladder has two literal abelian cohomological anchors: the inventory class $[\theta_R] \in H^1(\mathcal{S}; A)$ at nerve level two (a transported-resource datum) and the weld class $[\omega] \in H^2(\mathcal{S}; \mathcal{D})$ at nerve level three (the splitting obstruction of an associative weld). They are not ordered by detection power; they are *independent*. The results are proved in [9].

Theorem 14 ([9])**.** *Let $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ and let R be a resource local system with abelian stock-group A. Global inventory is consistent — R admits a global section over the cover $\{\mathcal{C}, \mathcal{D}\}$ — iff $[\theta_R] = 0 \in H^1(\mathcal{S}; A)$. The matched-pair law plays no role: a family compatible with the $\mathcal{C}$- and $\mathcal{D}$-morphisms is a global section, because compatibility is closed under composition and $\mathcal{C} \cup \mathcal{D}$ generates $\mathcal{S}$.*

Theorem 15 (independence, [9])**.** *$[\theta_R] \in H^1(\mathcal{S}; A)$ and $[\omega] \in H^2(\mathcal{S}; \mathcal{D})$ are logically independent: both "$[\omega] = 0$ with $[\theta_R] \neq 0$" and "$[\omega] \neq 0$ with $[\theta_R] = 0$" occur. Moreover no family of homomorphisms carries either class to the other: by the two witnesses, either direction would have to send a zero class to a nonzero class.*

Example 16 (the two witnesses)**.** For the first direction, let $\mathcal{S}$ be the free category on the diamond quiver $a \rightrightarrows \{b, c\} \rightrightarrows d$ with distinct routes: directed-acyclic, so no non-identity invertibles, the weld coefficient system is trivial and $[\omega] = 0$ is forced; yet $B\mathcal{S}$ has first Betti number 1, so a transport system with nonzero loop holonomy gives $[\theta_R] \neq 0$. The second direction is *generic*: for *any* $\mathcal{S}$

with $[\omega] \neq 0$, the zero inventory system has $[\theta_R] = 0$. A concrete such $\mathcal{S}$ is the *rigid-twist category* — one object with endomorphism monoid a copy of $\mathbb{Z}/2$ generated by a self-inverse twist — whose weld class generates $H^2(\mathcal{S}; \mathbb{Z}/2)$. The substantive direction is thus the first; the second needs only the freedom to choose the resource.

Remark 17 (what $[\omega]$ is a class of)**.** Theorem 14 assumes $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ is *given* as a matched-pair product, in which case $[\omega] = 0$ automatically; yet Theorem 15 and Example 16 exhibit $[\omega] \neq 0$. The apparent clash is resolved by saying precisely what $[\omega]$ is a class of (Eilenberg–Mac Lane / Baues–Wirsching): $[\omega] \in H^2(\mathcal{S}; D)$ is a class of the *base* $\mathcal{S}$ with coefficient system D — the vertex-group natural system carried by the candidate normal factor, *not* the wide subcategory $\mathcal{D}$ of a presupposed factorization. It classifies an extension $D \rightarrowtail E_\omega \twoheadrightarrow \mathcal{S}$, and it is E_ω (equivalently, the candidate assembled weld) that splits or not. When $\mathcal{S}$ is already presented as $\mathcal{C} \bowtie \mathcal{D}$ the extension is split by construction, so $[\omega] = 0$; Example 16 takes a base *not* so presented. For the rigid-twist witness, $\mathcal{S} = B(\mathbb{Z}/2)$, $D = \mathbb{Z}/2$, and $E_\omega = B(\mathbb{Z}/4)$: the non-split extension of $\mathbb{Z}/2$ by $\mathbb{Z}/2$, whose class generates $H^2(\mathbb{Z}/2; \mathbb{Z}/2) \cong \mathbb{Z}/2$.

Remark 18 (different degrees do not by themselves mean independence)**.** It would be a mistake to argue independence from the bare fact that the two classes sit in different degrees. Standard degree-raising maps exist — the transgression $d_2 \colon H^1 \to H^2$ of a Lyndon–Hochschild–Serre spectral sequence, and Bockstein homomorphisms $H^1 \to H^2$ — so a priori a class could be carried up a degree. Independence here is the *content of the two witnesses* (Theorem 15), not a formal consequence of the degree gap. What the degree gap does explain is why no *isomorphism* could identify the two classes outright: for the free loop $\mathcal{S} = B\mathbb{N}$ (one loop; $B\mathcal{S} \simeq S^1$) one has $H^1(B\mathbb{N}; A) = A$ while $H^2(B\mathbb{N}; A) = 0$, and for a weld coefficient one takes $\mathcal{D} = \mathbb{Z}$ with $H^2(B\mathbb{N}; \mathbb{Z}) = 0$ — the degrees genuinely do not match. This is the same "rhyme, not identity" caution that governs the two H^2-sites elsewhere in the programme [12], here across one degree: the same cyclic group can appear on both rungs, but the site, the coefficients, and the degree differ, and one must check all three before identifying two obstructions.

The conjecture tested in [9] — that inventory consistency *equals* the weld $[\omega] = 0$ — was a false generalization of blockchain re-entrancy, where the conserved quantity *is* the weld-state and no separate resource presheaf exists. In a supply chain the stock lives on the nodes, transported by operations, and is a genuinely separate degree-one datum. The modelling is sharpened, not debunked: re-entrancy is a nerve-level-three failure (the weld is associative but does not split — the hand-off cannot be cleanly factored), inventory inconsistency a nerve-level-two failure (route discrepancy, on the nodes) — two economic failure modes at two levels, each with a witness.

7 The program, and what it would take to prove

What is assembled here. Three invariant instances fall on a single nerve-level ladder: out-degree homogeneity at level one, blind to composition; the δ-translation invariant at level two, which detects groupoids and thin groupoids and separates $\mathbb{Z}/2$ from $\{1, e\}$ — at cohomological degree one; and the weld at level three, the splitting obstruction of an associative weld. Two non-collapse facts make the ladder informative rather than decorative: level-one blindness (Theorem 3), and the independence of the two cohomological anchors (Theorem 15). A third, collapse, keeps us honest: Theorem 11 shows there is no third *detection* rung for a fixed category.

The program, stated as a program.

The nerve-level program. For a directed container (small category) $\mathcal{C}$ and a compositional invariant I, the *nerve level* of the binding datum I reads records what I can detect: an invariant factoring through the k-truncation $\mathrm{tr}_k\mathrm{N}(\mathcal{C})$ is blind to every composition phenomenon genuinely of nerve level $> k$. In particular composition — the comultiplication δ, the composition map d_1 — is a nerve-level-two datum, first detectable at cohomological degree one; and associativity — the condition on composable triples that is the defining cocycle condition of the weld $[\omega] \in H^2$ (the class itself measuring splitting) — is the nerve-level-three datum, cohomological degree two.

This is a **conjecture and research direction, not a theorem**. We have a regrade lemma, a strict floor separation, a collapse theorem, an assembly separation, three instances and one independence fact; we do not have a uniform "blindness" theorem that an invariant factoring through tr_k cannot separate two categories agreeing to level k, nor a converse realizing every genuine level-k distinction. Note Theorem 11: for *isomorphism*-invariants the truncation grade is genuine only up to level two, so any such program must grade *conditions* (or assembly obstructions), not fixed-category truncations, above that line.

The open question, honestly. Does cohomological *degree* stratify *detection* power above the floor? **No, as naively stated**, and this is the heart of the correction. Theorem 7 exhibits a degree-one invariant (over $\mathbb{F}_2$) that detects composition, so "nothing below degree two detects composition" is false; and Theorem 15 shows the degree-one and degree-two anchors are independent, hence not ordered. What *is* true is the condition-level strictness of Theorem 12 (composition exists at N_2, is associative at N_3) and the coefficient-test of Remark 8 (the nerve level of the binding *datum* is coefficient-free; the degree, and the nerve level of the detecting class, are not). The grade that stratifies is nerve level; whether degree does anything beyond shadowing it is open.

The sharpened applications story. As a grant-narrative frame the ladder is immediately useful, and it sharpens rather than replaces the earlier picture. Compositional failure is not one phenomenon but several, cleanly separated by the nerve level of the conserved quantity:

Datum lives…	Nerve level	Failure class	Economic instance
on the **graph** (out-degrees)	1	invisible to ∂	tangent/shape geometry
on the **nodes** (transported)	2	$[\theta_R] \in H^1$ — route discrepancy	supply-chain inventory
in the **assembly** (weld)	3	$[\omega] \in H^2$ — weld does not split	blockchain re-entrancy

In our stylized transport model — stock carried by translations, not the flow-conservation and capacity constraints of operations-research inventory accounting — supply chains give an instance of "failure = nonzero H^1". The mechanism is classical: a route discrepancy is the obstruction to a node-potential, exactly Kirchhoff's voltage law, and the same coboundary condition governs arbitrage-freeness and the ranking inconsistencies measured by HodgeRank [5] and by sheaf-theoretic data fusion [6]. We therefore do *not* claim supply chains as the "first" such instance — that would need a literature search we have not run — only as a clean one in this programme, standing beside re-entrancy's H^2 at a different nerve level.

Acknowledgement and honesty. Our port of Lanfranchi's theorem [4] and our supply-chain conjecture tested in [9] are both sharpened here, not debunked: the first because its home is a Cartesian tangent category and the container derivative's is not, the second because inventory is a node-level degree-one datum and not the weld. The parallelizability definition the port leans

on is due to Cruttwell, Ikonicoff, Lemay and Van Der Linden, as Lanfranchi records; the port itself is ours. The nerve-level program is a heuristic organizing proved facts; its general form is open, and the slogan it replaces — "composition is a degree-two datum" — is retired. Every level assignment in this note tracks a datum we have computed, and the corrected slogan, "composition is a nerve-level-two datum, first seen at cohomological degree one," is underwritten by Lemma 1 and Theorems 5, 7 and 9.

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